

RAVI LAKKAPPA BELLUBBI

✉ ravi.lakkappabellubbi@gwu.edu ☎ 2027252985 🌐 ravi l bellubbi 🏠 Ravi L Bellubbi 📧 ravilbellubbi 📍 Washington, DC - 20006

🎓 EDUCATION

Master of Science - Computer Engineering (Specialization: VLSI & Computer Architecture); GPA : 3.57
The George Washington University

08/2023 – 05/2025
Washington DC, USA

Bachelor of Engineering - Electronics and Communication; CGPA : 7.73, Major GPA : 8.54
Dayananda Sagar College of Engineering

2018 – 2022
Bangalore, India

🧠 SKILLS

OS, EDA, FPGA & CPLD Tools

Linux, Cadence - Xcelium, Virtuoso, Voltus & Innovus, Tetramax, QuestaSim, Makerchip, MPLAB X IDE and IPE, Cheetah SPI host, Quartus prime, Synopsys VCS, WinMIPS64, Dinero, Scale-sim, Booksim2, Gem5

Hardware, Scripting & Assembly Languages

C, C++, C#, Verilog, TL-Verilog, System Verilog, VHDL, Nios ii, MATLAB, Simulink, Python, Perl, TCL, Bash, Makefile, Git, Git version control and Assembly language

Protocols, Peripheral Controllers and Interfaces

Ethernet, JESD204B, PCI, PCIe, UART, I2C, SPI, APB, AHB, AXI

System Architectures

ASIC, FPGA, Memory, SoC, CPU, GPU, O-RAN

📁 WORK EXPERIENCE

Trainee Intern - FPGA Design and Verification Engineer

Akkodis

06/2024 – 08/2024
Kings of Prussia, USA

- Verified PCIe IP components (DLL Transmitter/Receiver, Replay Buffer etc.) using UVM testbenches in QuestaSim, while optimizing low-power RTL coding for the PCIe DLL layer and assisting with synthesis, timing closure, and FPGA prototyping.
- Developed formal verification methods for AHB2APB bridge and contributed to chip-level verification.
- Performed FPGA design verification using Cadence Innovus, generated coverage reports in QuestaSim, conducted ATGP with Synopsys VCS, and automated testbench setup using Perl/Python, makefile scripting improving test efficiency.

Trainee Intern - Verification Engineer

Velodyne Lidar India Pvt. Ltd

07/2022 – 02/2023
Bengaluru, India

- Debugged Velarray M1600 by analyzing waveforms and understanding real-time testbench programming.
- Verified the scanner block in the Condor project by integrating MATLAB functions with SystemVerilog DPI-C, utilizing the Octave library in C/C++.
- Tested Lidar peripherals (top & bottom board) in the Next-Generation Platform project, gaining experience with MPLAB X IDE, IPE, Cheetah SPI host adapter, and Saleae Logic Analyzer.

Trainee - Advanced VLSI Design & Verification

Maven Silicon

02/2022 – 08/2023
Bengaluru, India

- Excelled skills in the domains of DFT, ASIC, FPGA Design & Verification Methodologies such as UVM-OVM and Assertion based Verification: SVA. Developed projects on Router 1x3 and AHB2APB Bridge Verification.

📁 PROJECTS

AES Encryption Core

The George Washington University

03/2025

- Designed and implemented an AES-128 encryption core in Verilog, utilizing a 10-round pipelined architecture with SubBytes, ShiftRows, MixColumns, and AddRoundKey transformations for high-throughput encryption.
- Optimized FPGA resource utilization through register retiming and logic balancing, improving area and power efficiency.
- Performed synthesis, back-annotation, and layout analysis, generating STA reports for timing closure, area, and clock performance.
- Verified the design using Tmax ATPG for fault coverage analysis, conducted functional and gate-level simulations, and generated coverage reports to validate correctness.

Machine Learning for Optimized Computer Architecture Designs

The George Washington University

12/2024

- Adjusted computer architecture parameters such as cache size, core frequency, voltage etc. for better performance, power efficiency, and reliability for CNN algorithm Accelerator
- Utilized Aladdin Gem5 and Smaug to evaluate area, power, execution time, energy consumption, and reliability, with my proposed architecture outperforming traditional heuristics by adapting to workloads.

Design of 256 Bit SRAM Memory Array

The George Washington University

12/2023

- Designed a low power 256 bit SRAM memory cell which is implemented using Cadence Virtuoso tool.
- Performed synthesis, back-annotation, and layout, generating reports for timing, area, and clock performance. Verified the design using Tmax tool and generated coverage reports
- Layout design is aligned into 90um padframe and generated a ready to use fabrication GDS file.

64-QAM modulator

The George Washington University

12/2023

- Implemented a 64-bit QAM Modulator with SPI, Symbol Mapping, CDC FIFO, and Data Register blocks, capable of processing a 60 Mbit/sec datastream and storing data at 10 MHz.
- Performed synthesis, back-annotation, and layout, generating reports for timing, area, and clock performance. Verified the design using Tmax tool and generated coverage reports

Design of a Radiation Hardened SRAM Memory Architecture

Dayananda Sagar College of Engineering

07/2022

- Designed a Radiation Hardened by Design (RHBD) SRAM cell with other axillary designs which are resilient to all soft errors caused by exposure to very high radiations at outer space.
- Developed & Verified the Design Layout using Cadence virtuoso & Innovus. Simulations are performed on LTSpice tool